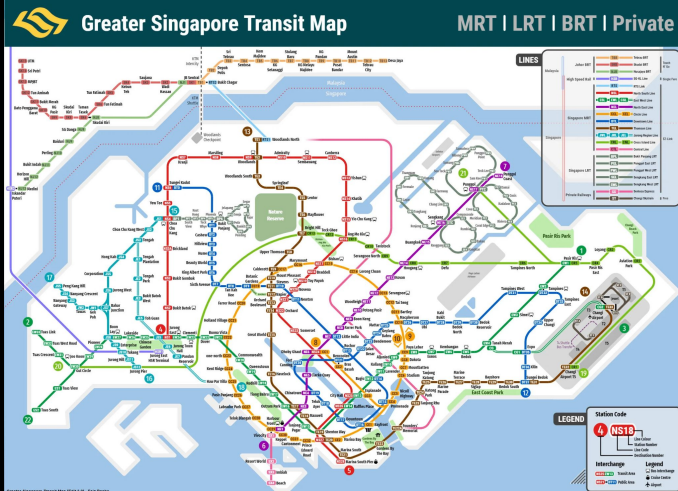




DECEMBER 12

Proximity effects on Singapore's real estate



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Motivation

- Firsthand Experience with Singapore's MRT System
- Challenges of Urban Planning in Dense Cities
- Complex Trade-Offs of Proximity to Transit
- Exploring Cross-Cultural Urban Dynamics





Goal

Determine the relationship between the proximity of the MRT on the flat values.

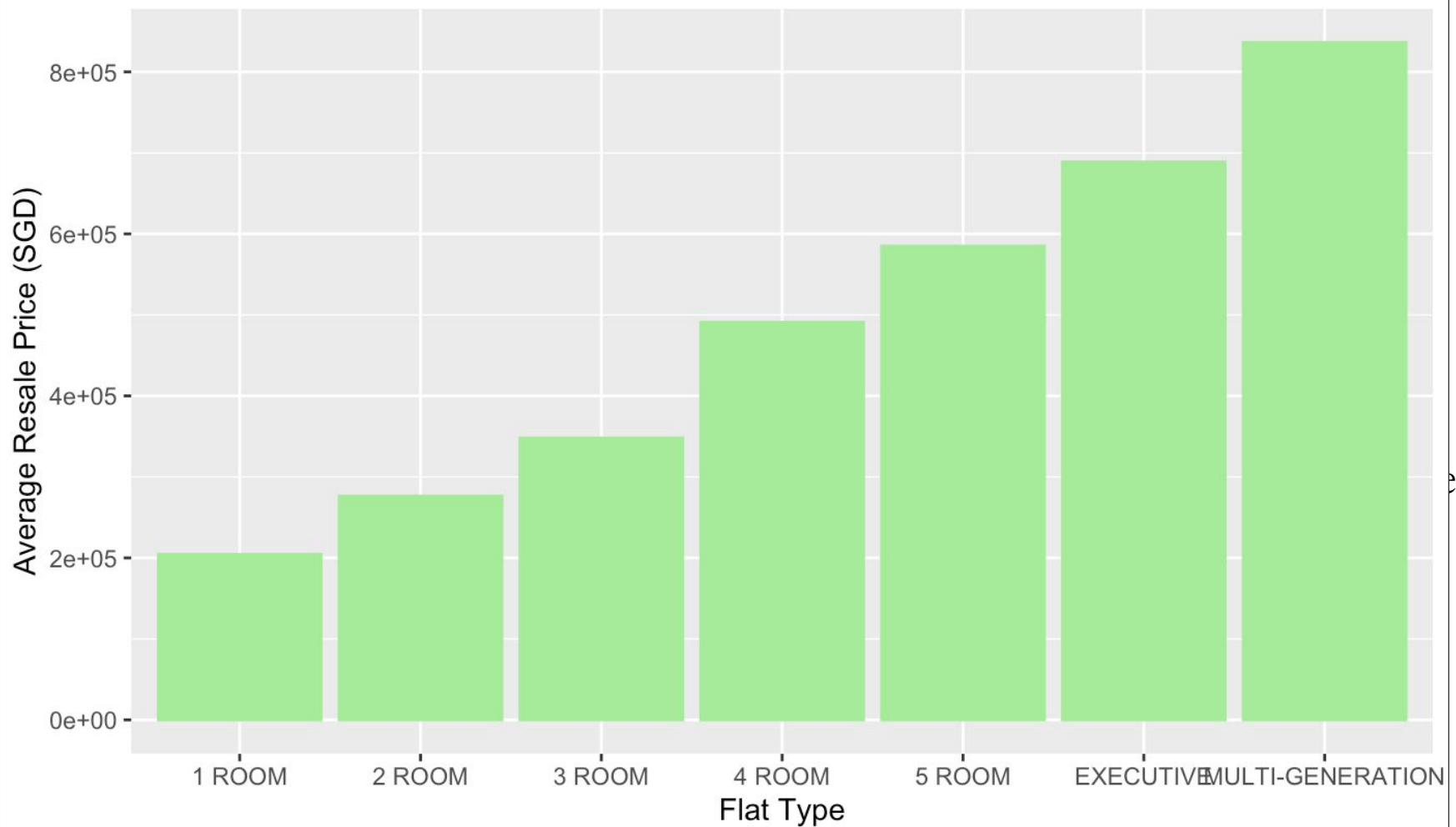
Data

- These datasets collectively provide information on 220,971 records of HDB resale prices (SGD) between 2015 and 2024, including details such as floor levels and distances to the MRT (meters)
- Sources: Singapore Housing Development Board (HDB), National Map Line (SLA) & OneMap Reverse Geocode
- Flat: Singapore's housing in the dataset
- Storey Range: the level that the flat is categorized as
- Expressway: the Mass Rapid Transportation (MRT)

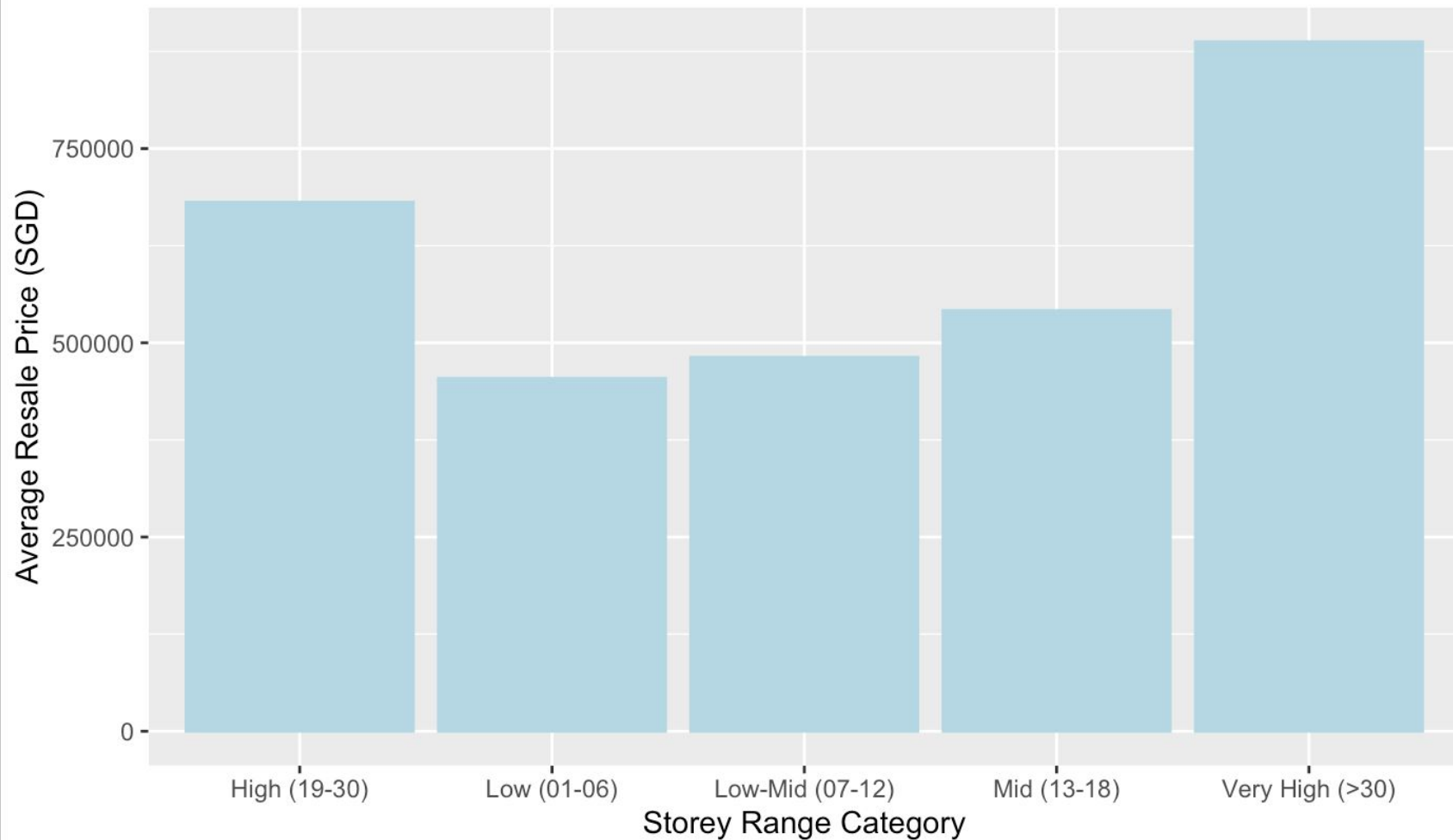
Data Limitations

- Exclusion of Socio-Economic Status
 - Salary Median
 - Preference of private vs. public transportation
- Price Variations from 2015–2024
 - Resale prices are not separated by year due to Singapore's relatively stable housing market, which might obscure year-specific trends
 - Singapore's 1.39% vs. U.S. 4.3% annual housing prices

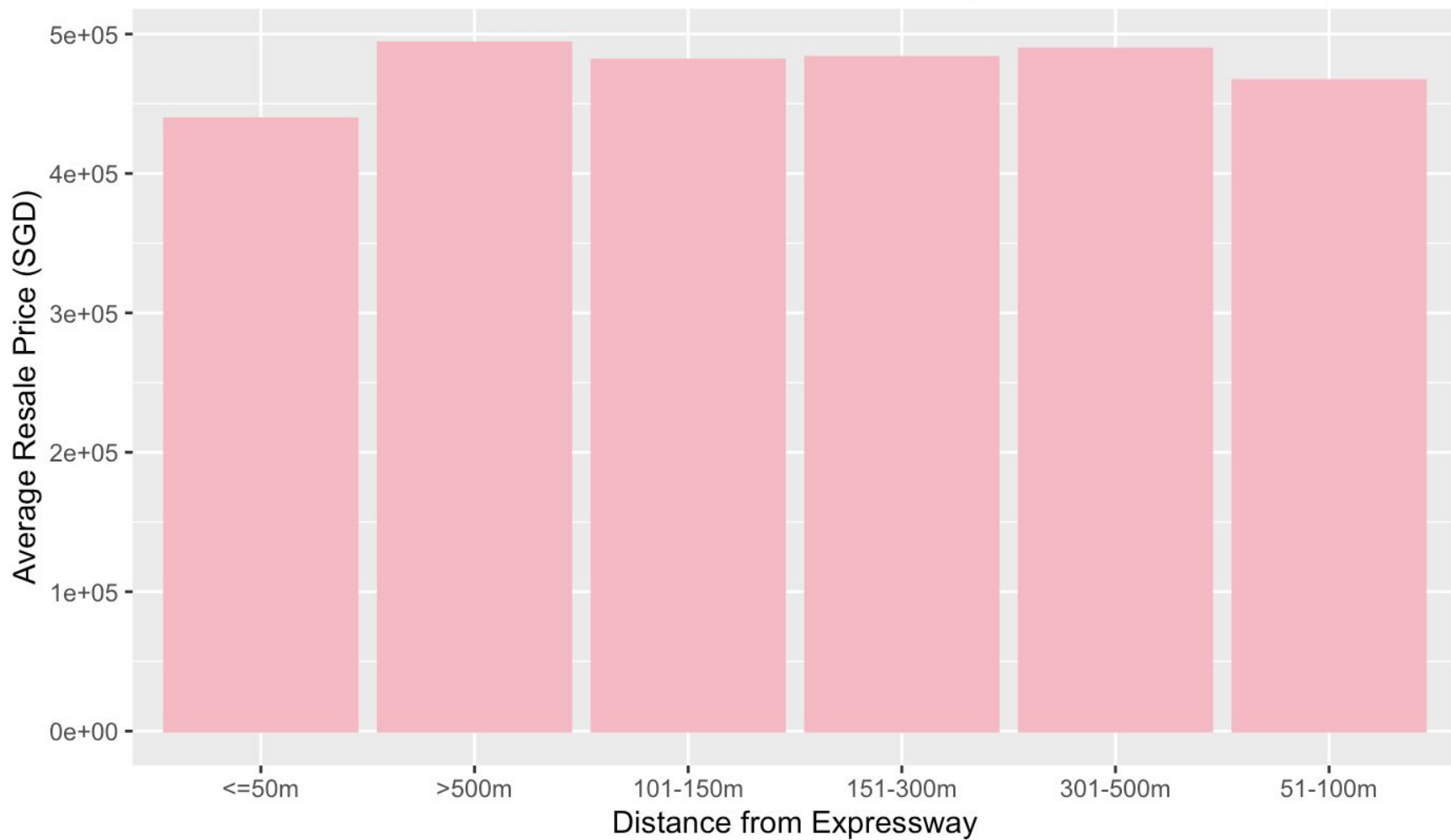
Average Resale Price by Flat Type



Average Resale Price by Storey Range Category



Average Resale Price by Distance from Expressway



Predictions

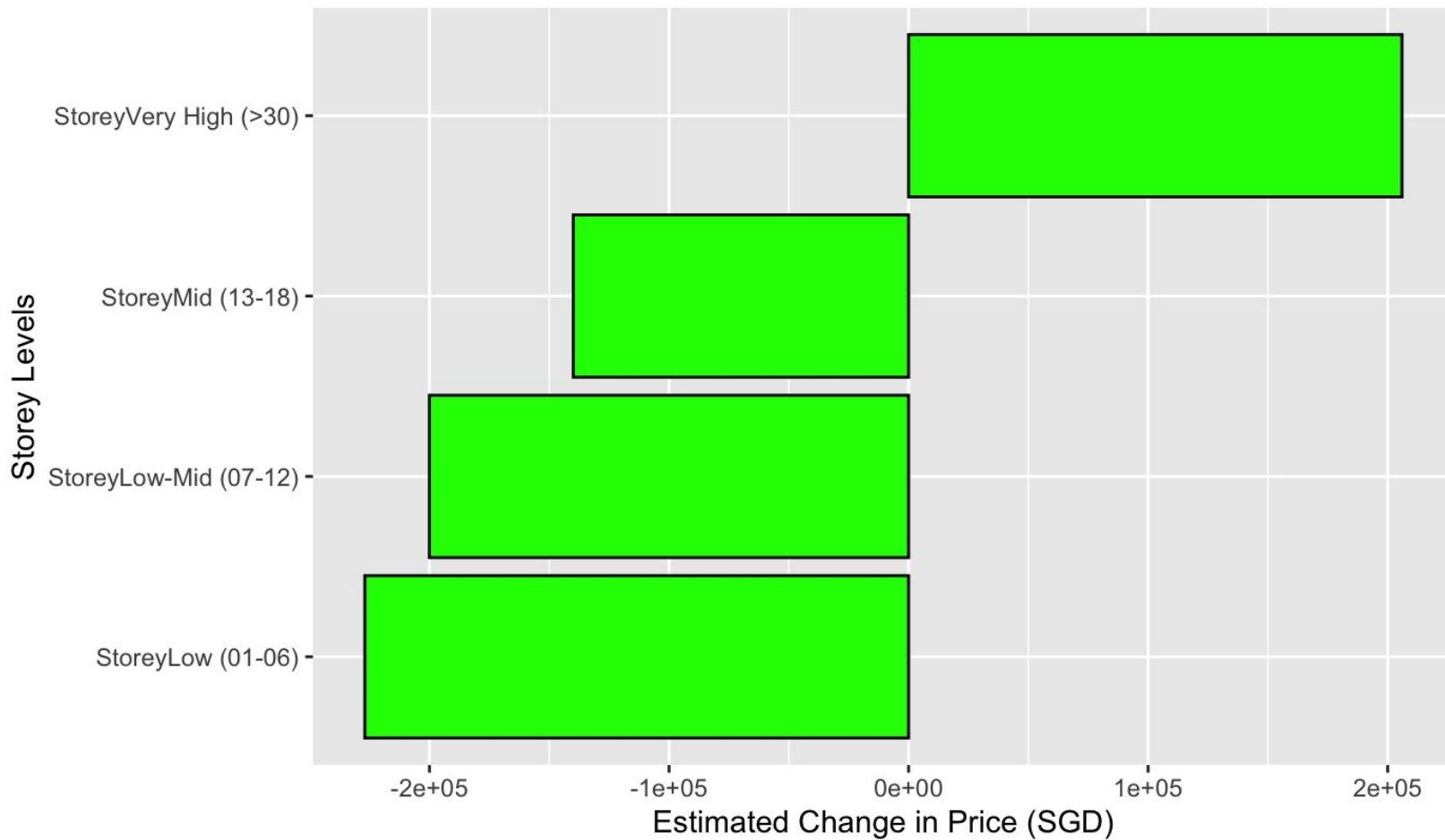
- Flats that are further away from the expressway has a higher resale price due to the benefit of having more privacy, noise pollution and different lifestyle.
- Flats that are higher in storey levels has a higher resale price due to better views, social status and pollution.



Analysis

Are these difference in resale prices significant?

Estimated Effects on Resale Price Based on Storey Levels





Storey Levels

Reaffirms prediction:

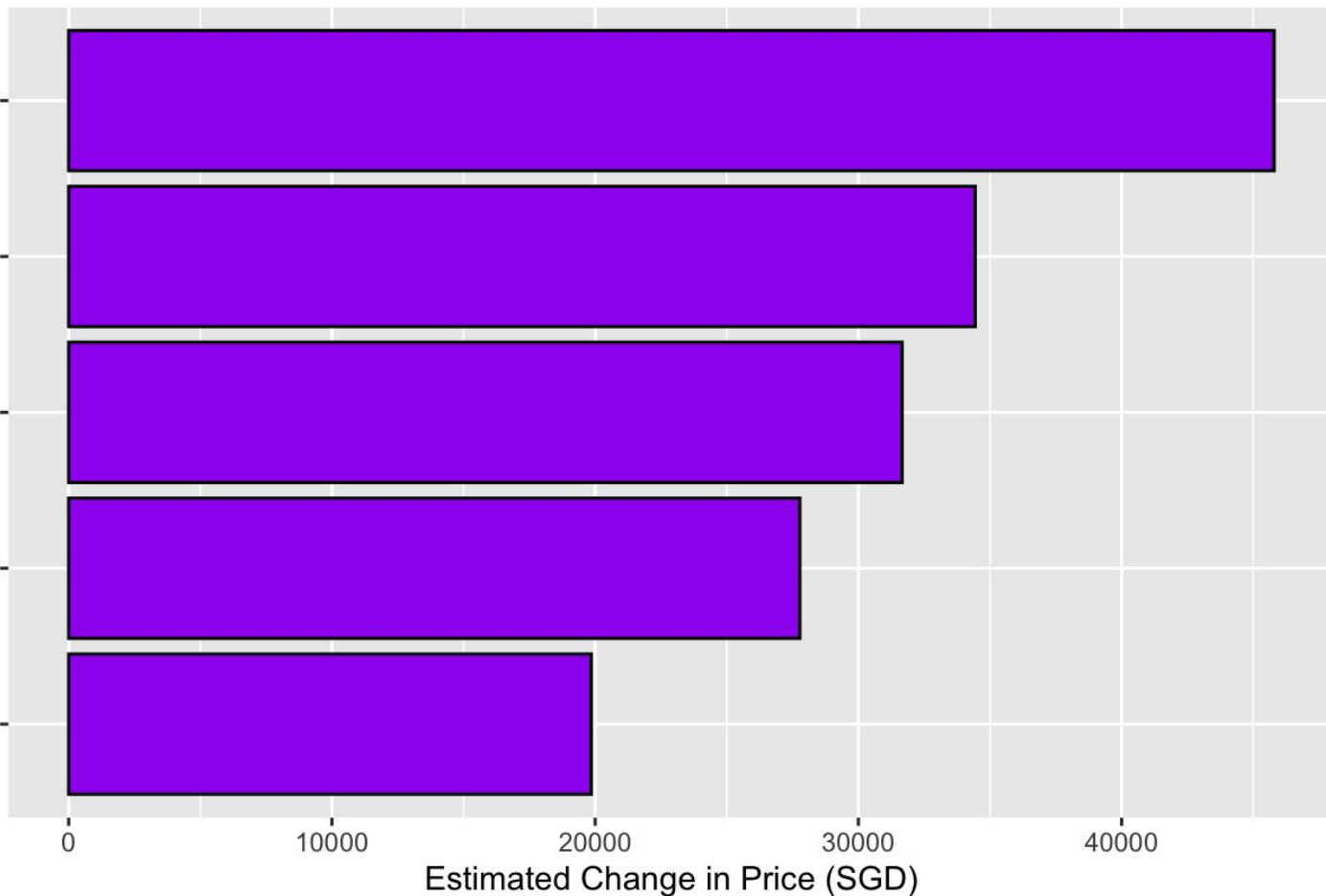
- Flats on higher floors has a higher resale value

Estimated Effects on Resale Price Based on Expressway Proximity

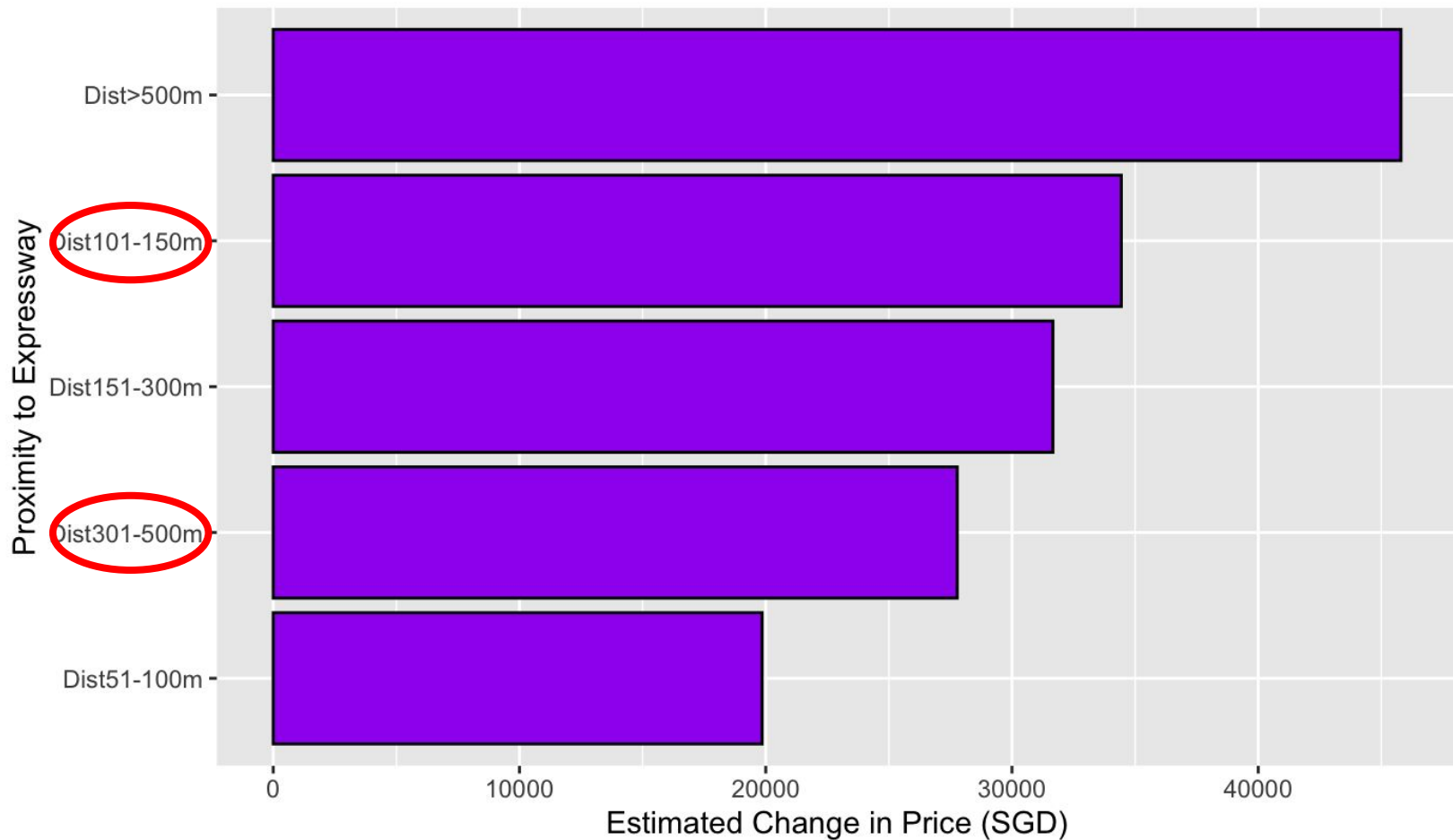


Proximity to Expressway

Dist>500m
Dist101-150m
Dist151-300m
Dist301-500m
Dist51-100m



Estimated Effects on Resale Price Based on Expressway Proximity





Proximity

High distance ($>500\text{m}$) & Low Middle distance (100–150m)
flats away from the express way are higher valued.

Significance

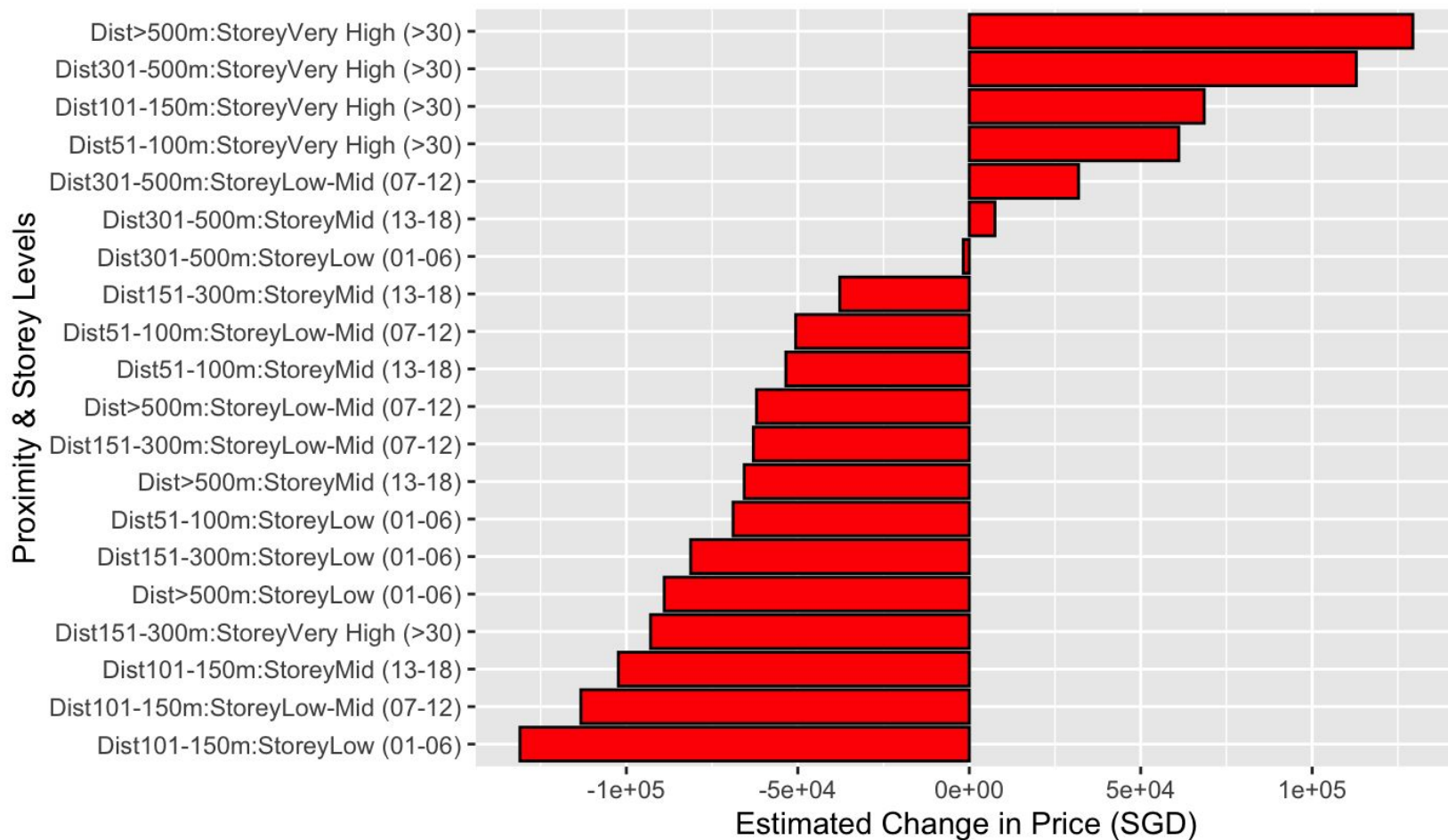


	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	638337	3085	206.940	< 2e-16	***
distance_from_expressway>500m	45795	2666	17.176	< 2e-16	***
distance_from_expressway101-150m	34440	3309	10.409	< 2e-16	***
distance_from_expressway151-300m	31663	3189	9.927	< 2e-16	***
distance_from_expressway301-500m	27777	3650	7.610	2.75e-14	***
distance_from_expressway51-100m	19854	3143	6.316	2.69e-10	***
storey_range_categoryLow (01-06)	-227007	1679	-135.205	< 2e-16	***
storey_range_categoryLow-Mid (07-12)	-200040	1682	-118.963	< 2e-16	***
storey_range_categoryMid (13-18)	-139980	1835	-76.290	< 2e-16	***
storey_range_categoryVery High (>30)	205962	3920	52.536	< 2e-16	***

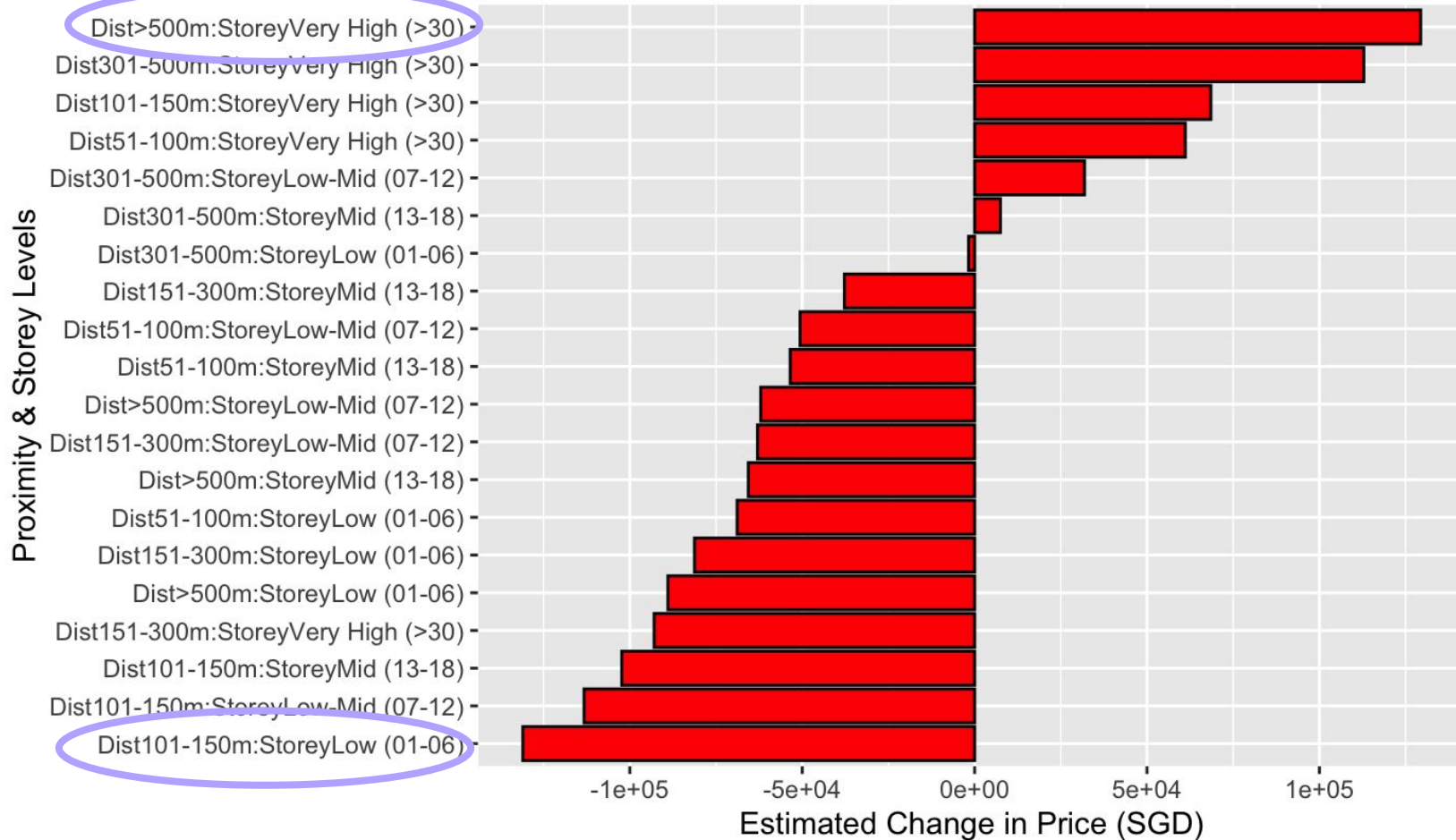


Relationship Between Storey Levels and Expressway Proximity

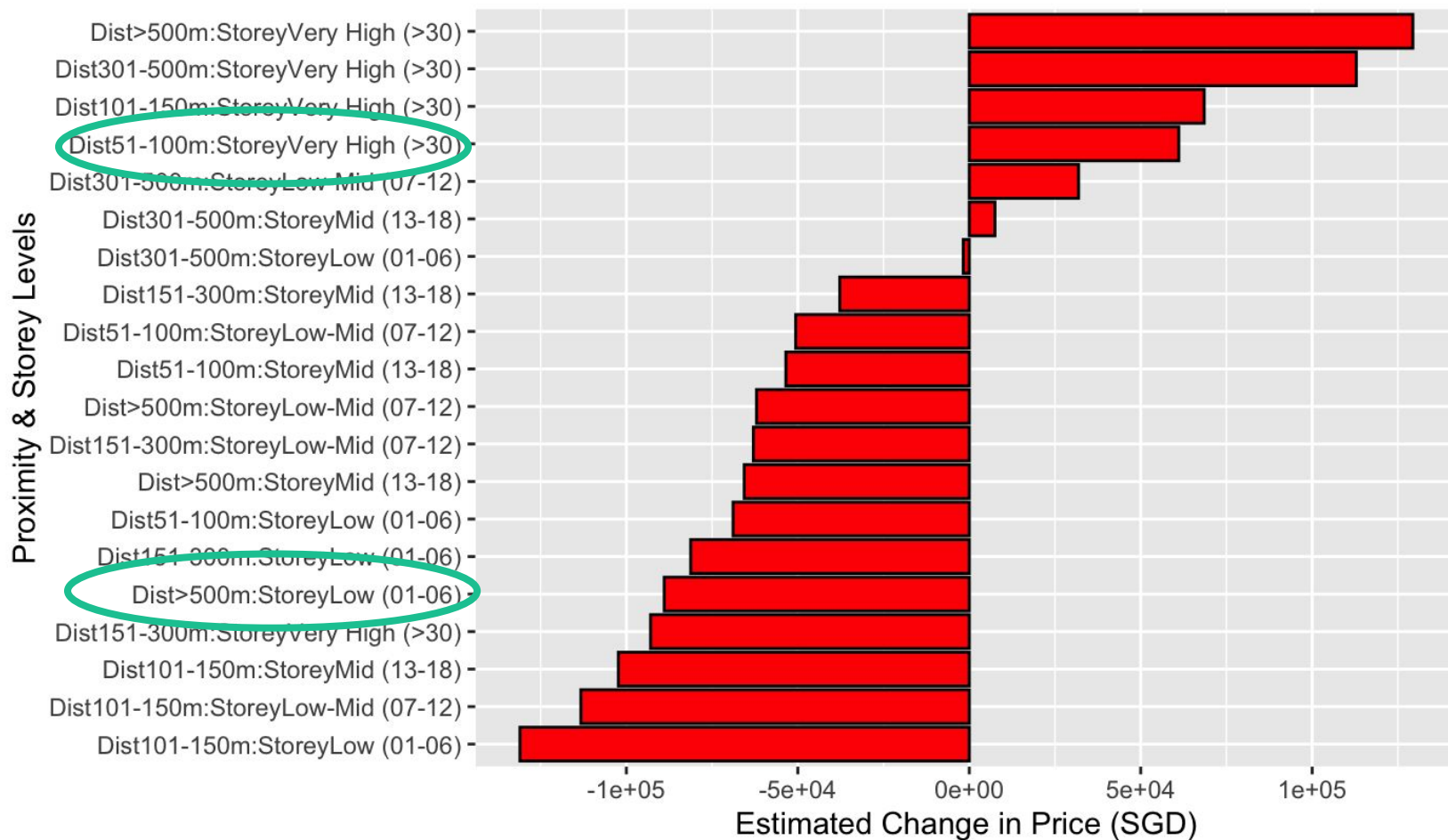
Expressway Proximity & Storey Levels Effects



Expressway Proximity & Storey Levels Effects



Expressway Proximity & Storey Levels Effects





Results

- Storey Range has more impact on the price of a Singaporean flat
- The extremes stays true, higher and further flats are expensive, and lower and closer flats are more affordable.
- Low Middle Distance (100m–150m) is a preferred distance.

In Summary



The interaction between expressway proximity and storey levels further highlights that housing prices are influenced by the proximity from people's major mode of transportation and amenity.

These findings underscore the importance of considering both proximity to infrastructure and housing design when evaluating property values.

Singaporean housing markets and its public transportation relationships may be applicable to U.S. dense cities to human's preference to luxury, convenience and privacy.



Future

- Additional variables, e.g. median salary, noise levels and proximity to other amenities.
- Categorize prices in groups, reducing the range for a fairer comparison.